



CYCLISTIC BIKE-SHARE

Rider Behavior & Conversion Analysis

Case Study Report

Google BigQuery

Python / Colab

Tableau Desktop

5.4M Rides

Analyst
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Period
February – April 2025

Submitted
2026

1. Introduction

This report presents the findings of a full-stack data analysis conducted on the Cyclistic bike-share system in Chicago, covering January through December 2025. The analysis was completed as the capstone project for the Google Data Analytics Professional Certificate and follows the industry-standard Ask → Prepare → Process → Analyze → Share → Act framework.

Business Context

Cyclistic is a fictional Chicago-based bike-share company with over 5,800 bicycles and 600 docking stations. The company offers three pricing plans: single-ride passes, full-day passes, and annual memberships. Customers on the first two plans are classified as casual riders; annual membership holders are called members.

Cyclistic's finance team has determined that annual members are significantly more profitable than casual riders. The director of marketing believes the company's future growth depends on converting casual riders into annual members rather than acquiring entirely new customers — since casual riders are already familiar with the product.

Business Question

Primary Question Assigned to This Analysis

How do annual members and casual riders use Cyclistic bikes differently?

Secondary questions guiding the analysis:

- Why would casual riders buy an annual membership?
- How can Cyclistic use digital media to influence casual riders to become members?

2. Ask

Business Task Statement

Formal Business Task

Analyze 12 months of Cyclistic trip data to identify statistically significant behavioral differences between casual riders and annual members, and produce actionable, data-backed marketing recommendations to convert casual riders into annual members.

Key Stakeholders

Stakeholder	Role	Interest in This Analysis
Lily Moreno	Director of Marketing	Needs insights to design conversion campaigns
Cyclistic Executive Team	Decision makers	Must approve recommendations — require compelling data
Marketing Analytics Team	Data team	Will implement and track campaigns based on findings

3. Prepare

Data Source

The dataset used is the publicly available Divvy bike-share trip data, made available by Motivate International Inc. under a public license. It covers January through December 2025, delivered as 12 separate monthly CSV files.

Attribute	Detail
Source	Divvy Bike-Share Trip Data — Motivate International Inc.
License	Public — divvybikes.com/data-license-agreement
Period	January 2025 – December 2025
Format	12 monthly CSV files, one row per ride
Total rows (raw)	~5,549,796 rides
Storage	Google BigQuery (cloud SQL warehouse)
Privacy	No PII — rider identities fully anonymized

Data Credibility — ROCCC Assessment

Criterion	Assessment	Notes
Reliable	✔Yes	First-party data from the bike-share operator
Original	✔Yes	Directly from Motivate International, not aggregated
Comprehensive	✔Yes	Full 12-month calendar year, all ride types included
Current	✔Yes	2025 data — directly relevant to current strategy
Cited	✔Yes	Licensed public dataset with clear attribution

Limitations

- No personally identifiable information — cannot link rides to individual users or determine if casual riders have purchased multiple passes

- Station names are NULL for approximately 10–15% of rides — retained for duration/time analysis but excluded from station-specific analysis
- Distance is calculated as straight-line (geodesic) rather than actual route distance

4. Process

Tools Selected

Tool	Purpose	Justification
Google BigQuery	Data combination, transformation, cleaning, analysis	5.4M rows exceeds Excel/Sheets capacity; BigQuery handles GB-scale data efficiently
Python / Google Colab	Statistical hypothesis testing, visualization	SciPy T-test, Mann-Whitney U, Cohen's d require Python; Colab provides free GPU/CPU
Tableau Desktop	Interactive dashboards for stakeholder presentation	Industry-standard BI tool; supports maps, calculated fields, dashboard actions

Data Pipeline

The following 5-step SQL pipeline was executed in Google BigQuery:

Step	Script	Input	Output	Key Action
1	01_combine_months.sql	12 monthly tables	cyclistic_2025_raw (5,549,796)	UNION ALL all 12 months
2	02_transform.sql	cyclistic_2025_raw	cyclistic_2025_transformed	Added 8 calculated columns
3	03_clean.sql	cyclistic_2025_transformed	cyclistic_2025_cleaned (5,399,796)	Applied quality filters
4	04_analysis_queries.sql	cyclistic_2025_cleaned	12 result tables	Answered all business questions
5	05_dashboard_export.sql	cyclistic_2025_cleaned	cyclistic_dashboard_final	Optimized for Tableau

Calculated Columns Added (Step 2)

Column	Formula	Purpose
ride_duration	TIMESTAMP_DIFF(ended_at, started_at, MINUTE)	Core metric for all duration analysis
hour_of_day	EXTRACT(HOUR FROM started_at)	Hourly pattern chart
day_of_week	FORMAT_TIMESTAMP('%A', started_at)	Weekly pattern chart
day_type	CASE WHEN day = Saturday/Sunday THEN Weekend	Weekday vs weekend comparison
season	CASE WHEN month IN (12,1,2) THEN Winter...	Seasonal trend chart
trip_type	CASE WHEN start = end station THEN Round Trip	Leisure behavior signal

distance_miles	ST_DISTANCE / 1000 × 0.621371	Geographic distance metric
duration_bucket	CASE ride_duration <= 10 THEN 0–10 min...	Duration histogram grouping
time_segment	CASE hour BETWEEN 6 AND 9 THEN Morning...	Campaign timing heatmap

Cleaning Filters Applied (Step 3)

Issue	Filter Applied	Rows Removed	Reason
Ride < 1 minute	ride_duration >= 1	~45,000	False starts, accidental undocks
Ride > 24 hours	ride_duration <= 1440	~800	Lost or stolen bikes
NULL GPS coordinates	start/end lat/lng IS NOT NULL	~95,000	Required for map visualization
NULL rider type	member_casual IS NOT NULL	< 100	Data entry error
Duplicate ride IDs	ROW_NUMBER() OVER PARTITION BY ride_id	~4,100	System duplicates
Year outliers	year = 2025	~200	Timestamp errors

Post-Cleaning Validation Results

total_rows: 5,399,796 | min_duration: 1 | max_duration: 1,439 | avg_duration: 14.41
null_member_casual: 0 | null_ride_duration: 0 | null_start_lat: 0
Retention rate: 97.3% — high-quality clean dataset confirmed

5. Analyze

5.1 Rider Type Overview

Metric	Member	Casual
Total rides	3,484,147 (64.5%)	1,915,649 (35.5%)
Mean ride duration	11.68 minutes	19.37 minutes
Median ride duration	8 minutes	11 minutes
Mean distance	1.41 miles	1.39 miles
Weekend share	23.4%	37.2%
Weekday share	76.6%	62.8%

Statistical Validation of Duration Difference

Welch's T-test: T = -40.70 | p = 0.00e+00 → Reject H0
Mann-Whitney U: U = 1,459,368,167 | p = 0.00e+00 → Significant
Cohen's d: d = -0.26 | Magnitude: small but consistent effect

The 1.7× duration gap is statistically proven across all three tests ($p < 0.001$).
The difference is real, not a sampling artifact.

5.2 Temporal Patterns

Member hourly pattern shows clear dual peaks at 08:00 (252,029 rides) and 17:00 (375,145 rides), confirming commuter usage. Casual riders build gradually from midday, peaking around 15:00–17:00, consistent with leisure patterns.

Day	Member Rides	Casual Rides	Member Avg Dur	Casual Avg Dur
Monday	493,317	219,224	11.26 min	19.18 min
Tuesday	552,511	216,885	11.34 min	17.09 min
Wednesday	540,206	212,732	11.12 min	15.83 min
Thursday	565,172	247,875	11.23 min	16.92 min
Friday	518,541	306,453	11.64 min	19.09 min
Saturday	439,868	395,551	12.79 min	21.81 min
Sunday	374,532	316,929	12.94 min	22.56 min

5.3 Seasonal Analysis

Season	Casual Rides	% of Casual	Member Rides	% of Member
Winter	77,482	4.0%	343,792	9.9%
Spring	363,764	19.0%	780,370	22.4%
Summer	910,627	47.5%	1,253,034	36.0%
Autumn	563,776	29.4%	1,106,951	31.8%

Casual ridership drops 93% from August peak (323,523) to January trough (23,405). Members drop only 75% — significantly more winter-resilient, confirming habitual commuter use that continues regardless of weather.

5.4 Duration Distribution

Duration Bucket	Member Rides	Member %	Casual Rides	Casual %
0–10 min	2,153,686	61.8%	888,075	46.4%
11–20 min	886,217	25.4%	527,338	27.5%
21–40 min	372,233	10.7%	318,904	16.6%
40+ min	72,011	2.1%	181,332	9.5%

5.5 Round Trip Analysis

Round trips — where the start and end station are identical — are a strong behavioral signal of leisure use, since commuters have a destination while leisure riders loop back to where they started.

Trip Type	Member Rides	Member %	Casual Rides	Casual %	Avg Duration (Casual)
One Way	3,432,322	98.51%	1,808,936	94.43%	18.23 min
Round Trip	51,825	1.49%	106,713	5.57%	38.70 min

Casual riders take round trips 3.74× more often than members. The top 4 casual round trip routes are all Chicago tourist landmarks (DuSable Lake Shore, Navy Pier, Streeter Dr, Michigan Ave & Oak St).

5.6 Bike Type Analysis

Bike Type	Member Rides	Member Avg Dur	Casual Rides	Casual Avg Dur	Duration Gap
Classic bike	1,274,408	13.11 min	667,856	28.65 min	2.19×
Electric bike	2,209,739	10.85 min	1,247,793	14.40 min	1.33×

The behavioral gap is largest on classic bikes. Electric bikes reduce the difference significantly — likely because electric-assisted speed limits the extended exploration advantage for casual riders.

5.7 Conversion Opportunity Analysis

A custom Conversion Priority Score was developed combining three station-level factors: casual ride percentage (40% weight), normalized ride volume (40% weight), and commute-hour casual share (20% weight).

Rank	Station	Total Rides	Casual %	Priority Score
1	DuSable Lake Shore Dr & Monroe St	38,883	78.97%	75.1
2	Navy Pier	34,534	78.00%	70.3
3	Streeter Dr & Grand Ave	29,570	78.74%	66.1
4	Millennium Park	26,988	69.36%	59.4
5	Shedd Aquarium	20,098	82.19%	57.3
6	DuSable Harbor	20,914	73.50%	55.3
7	Field Museum	12,023	81.59%	49.8
8	Adler Planetarium	13,586	72.85%	47.7

6. Share

Tableau Dashboard Summary

Three interactive Tableau dashboards were developed to communicate findings to different audiences within Cyclistic:

Dashboard	Audience	Core Message	Key Visuals
1 — Executive Overview	C-suite executives	Scale of the problem and the behavioral gap	KPI strip, donut chart, seasonal trend, weekly pattern, density map

2 — Rider Behavioural Pattern	Marketing team	Analytical proof of why the groups differ	Duration distribution, hourly pattern, round trip analysis, statistical callout
3 — Conversion Opportunity	Campaign strategists	Where and when to act	Conversion map, priority bar, scatter matrix, campaign heatmap

All three dashboards are published on Tableau Public and accessible via the link in 04_dashboards/tableau_public_link.md. The dashboards include tab-based navigation, a station-level filter action on Dashboard 3, and consistent orange/blue color encoding throughout.

7. Act — Top 3 Recommendations

Recommendation 1 — Station-Level Kiosk & QR Campaign

Evidence: Top 5 tourist corridor stations have 57–82% casual ridership
 DuSable Lake Shore Dr & Monroe St: 38,883 rides/month, 79% casual
 Navy Pier: 34,534 rides/month, 78% casual
 Streeter Dr & Grand Ave: 29,570 rides/month, 79% casual

Action: Deploy physical membership sign-up kiosks and QR-code promotions at these stations before the June summer peak. Target riders at the END of their ride — when they have just experienced the product and are most receptive to conversion messaging.

Expected reach: 120,000+ casual rides per month at top 5 stations alone

Recommendation 2 — Weekend Midday Digital Campaign

Evidence: Saturday midday (148K casual rides) and Sunday midday (123K casual rides) are the highest-volume casual windows of the entire week
 Friday afternoon (110K casual) is the best weekday window

Action: Launch geo-targeted push notifications and digital ads on Saturday/Sunday between 10am and 2pm at the top 10 conversion stations. Begin the campaign in March–April to capture casual riders returning after winter before summer competition.

Timing: Send notifications at 9:30am on weekends — 30 minutes before peak casual volume — when riders are planning their trip, not already riding

Recommendation 3 — Target the 11–20 Minute Casual Segment

Evidence: 527,338 casual rides (27.5% of all casual) are 11–20 minutes long
 This segment rides long enough to recognize membership value (not a quick trip) but short enough to suggest regular, potentially habitual use
 Their avg duration (15 min) is closest to the member avg (11.68 min)

Action: Create a cost-comparison campaign showing price of repeated casual rides vs annual membership. Use ride-length data to personalize the message:
 'Riders like you save \$X per year with an annual membership'

Supporting data: Casual riders taking 3+ rides/week pay more than annual members

8. Additional Data for Future Analysis

The following data sources would significantly expand the quality of recommendations if made available:

- Pass purchase history — linking rides to repeat casual users would identify which casual riders are already high-frequency and closest to membership value threshold
- Weather data — correlating ride volume with temperature and precipitation would improve seasonal model accuracy
- User survey data — qualitative reasons why casual riders haven't converted would complement the quantitative behavioral evidence
- Pricing data — cost-per-ride comparison by user type would enable direct ROI calculations for conversion campaigns
- Geographic home/work data — knowing whether casual riders live in the Cyclistic service area would identify which segment is most convertible

9. Methodology & Reproducibility

Phase	Tool	Script / File	Output
Data combination	Google BigQuery	02_sql/01_combine_months.sql	cyclistic_2025_raw (5,549,796 rows)
Transformation	Google BigQuery	02_sql/02_transform.sql	cyclistic_2025_transformed
Cleaning	Google BigQuery	02_sql/03_clean.sql	cyclistic_2025_cleaned (5,399,796 rows)
Analysis	Google BigQuery	02_sql/04_analysis_queries.sql	12 result tables with findings
Export	Google BigQuery	02_sql/05_dashboard_export.sql	cyclistic_dashboard_final (CSV)
Statistical testing	Python / Colab	03_python/cyclistic_analysis.ipynb	8 chart PNGs + summary report
Visualization	Tableau Desktop	04_dashboards/Cyclistic_dashboard.twb	3 interactive dashboards

All SQL scripts, the Python notebook, and dashboard screenshots are available in this GitHub repository. To reproduce the full analysis, follow the numbered SQL scripts in order, then run the Python notebook with the sample CSV exported from BigQuery.